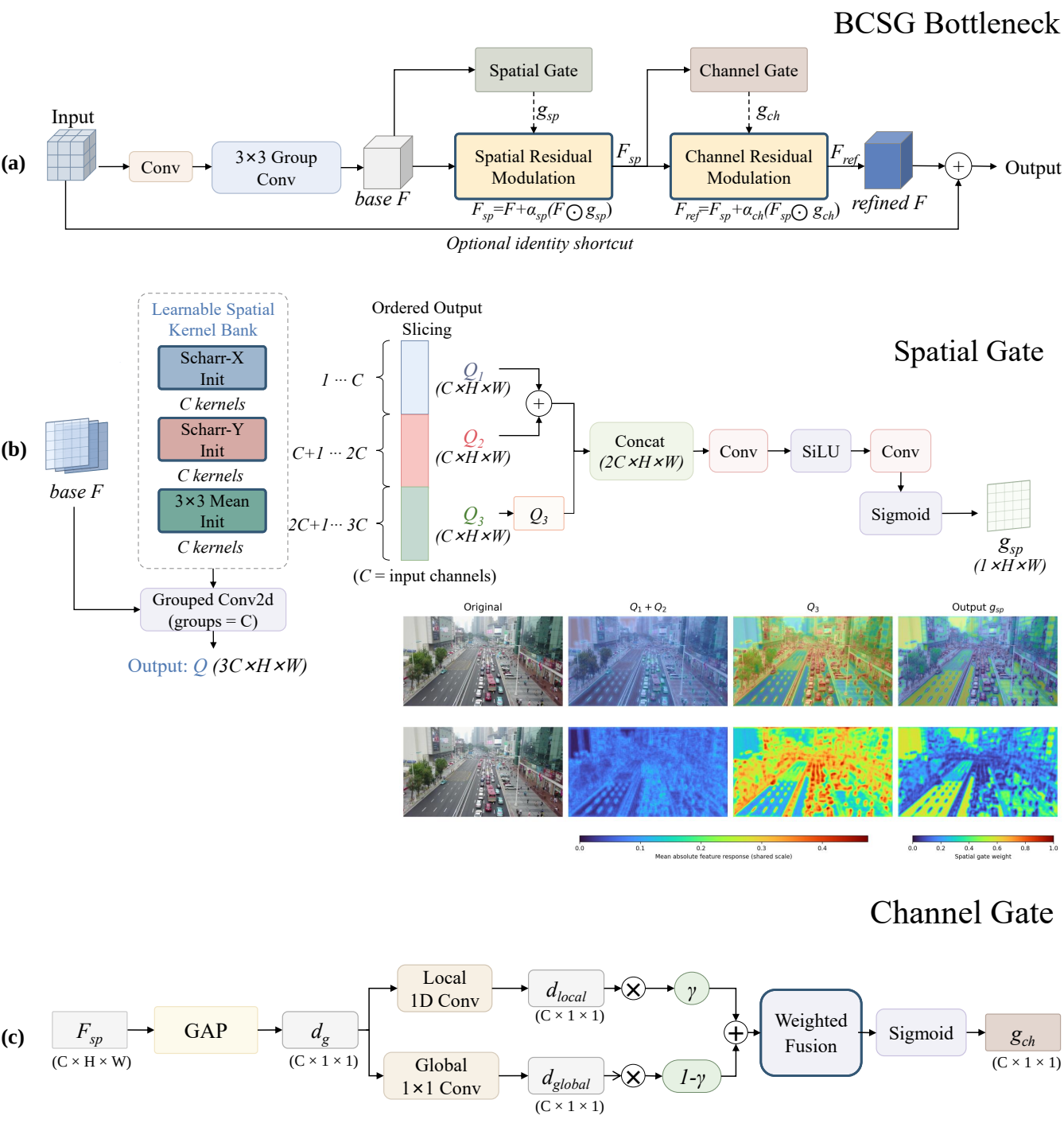

BCSG Bottleneck
(a)
Input
Conv
3×3 Group Conv
base F
Spatial Gate
g_{sp}
Spatial Residual Modulation
$F_{sp}=F+\alpha_{sp}(F\odot g_{sp})$
F_{sp}
Channel Gate
g_{ch}
Channel Residual Modulation
$F_{ref}=F_{sp}+\alpha_{ch}(F_{sp}\odot g_{ch})$
F_{ref}
refined F
Output
Optional identity shortcut
Spatial Gate
(b)
base F
Learnable Spatial Kernel Bank
Scharr-X Init
C kernels
Scharr-Y Init
C kernels
3×3 Mean Init
C kernels
Grouped Conv2d (groups = C)
Output: Q (3C×H×W)
Ordered Output Slicing
1 ··· C
C+1 ··· 2C
2C+1··· 3C
Q_1 (C×H×W)
Q_2 (C×H×W)
Q_3 (C×H×W)
Q_3
(C = input channels)
Concat (2C×H×W)
Conv
SiLU
Conv
Sigmoid
g_{sp} (1×H×W)
Original
Q_1+Q_2
Q_3
Output g_{sp}
Mean absolute feature response (shared scale)
Spatial gate weight
Channel Gate
(c)
F_{sp}
(C × H × W)
GAP
d_g
(C × 1 × 1)
Local 1D Conv
d_{local}
(C × 1 × 1)
Global 1×1 Conv
d_{global}
(C × 1 × 1)
γ
$1-\gamma$
Weighted Fusion
Sigmoid
g_{ch}
(C × 1 × 1)